Laptop Price

Project Presentation

Objective

- Predict Prices: Estimate the probable price of a laptop based on its technical specifications. This helps companies determine appropriate pricing for new products, stay competitive in the market, and price according to customer expectations.

- Identify Key Features: Analyze which features have the most impact on pricing. This helps balance cost and performance in product design. Businesses can focus on the features that are most valuable to customers.

- Market Segmentation: Divide laptops into categories such as Budget, Premium based on customers needs and purchasing power. This enables companies to target the right customer segments and market their products more effectively, leading to increased sales

Objective

- Anomaly Detection: Identify extremely cheap or expensive products that deviate from typical pricing trends. This plays a vital role in maintaining data quality, fixing listing errors, and preserving customer trust.

- Support Strategy: Use insights gained from data to make strategic decisions related to sales, inventory management, and product development. This helps businesses understand market demands, create better products, and gain a competitive advantage.

Scope

- Identification of Key Factors Influencing Pricing: This project analyzes which internal (such as technical specifications) and external (such as market trends) factors influence the price of a product. This information helps businesses decide which aspects they should focus on for better pricing decisions.
- Market Segmentation: Laptops are classified based on their features and prices to understand which products are suitable for which type of customers. This segmentation helps businesses reach their target customers more effectively.
- Identification of Pricing Anomalies: Some products are priced significantly higher or lower than the normal range, which may indicate data errors or listing mistakes. This project helps identify such cases so they can be corrected in time.
- Product Classification by Price Category: Laptops are categorized into price ranges such as Budget, Midrange, and Premium so that consumers can find suitable options based on their needs and budget, and businesses can position their products effectively in the market.

Technology

2

Excel

Version 19H2

4

Python

Version 3.13

1

Windows 11

Version 23H2

3

Jupyter

Version 7.3.2

Data Collection

1. Laptop Price

https://drive.google.com/file/d/1sU3TLwZA306lXScMvY_qgZs7-BvffSEw/view

2. Form of Data

- Data is in csv form having quantitative as well as qualitative data.

3. Rows & Columns

- Rows: 1275 and columns: 25

Data Cleaning & Pre-Processing

```
import pandas as pd
df = pd.read_csv(r"C:\Users\Public\Downloads\laptop_prices.csv", encoding='utf-8-sig')
df.head()
```

	Company	Product	TypeName	Inches	Ram	OS	Weight	Price_euros	Screen	ScreenW	...	RetinaDisplay	CPU_company	CPU_freq	CPU_model
0	Apple	MacBook Pro	Ultrabook	13.3	8	macOS	1.37	1339.69	Standard	2560	...	Yes	Intel	2.3	Core i5
1	Apple	Macbook Air	Ultrabook	13.3	8	macOS	1.34	898.94	Standard	1440	...	No	Intel	1.8	Core i5
2	HP	250 G6	Notebook	15.6	8	No OS	1.86	575.00	Full HD	1920	...	No	Intel	2.5	Core i5 7200U
3	Apple	MacBook Pro	Ultrabook	15.4	16	macOS	1.83	2537.45	Standard	2880	...	Yes	Intel	2.7	Core i7
4	Apple	MacBook Pro	Ultrabook	13.3	8	macOS	1.37	1803.60	Standard	2560	...	Yes	Intel	3.1	Core i5

5 rows × 23 columns

We used python library pandas. We import csv file into (df) dataFrame. Using read_csv method.

Data Cleaning & Pre-Processing

- Clean and Convert Ram from String to Integer

```
df['Ram'] = df['Ram'].astype(str).str.replace('GB', '').astype(int)
```

Removes the 'GB' text from RAM values and converts the result to integer (e.g., '8GB' → 8).

- Clean and Convert Weight from String to Float

```
df['Weight'] = df['Weight'].astype(str).str.replace('kg', '').astype(float)
```

Strips 'kg' from strings like '2.2kg' → '2.2', and converts them into float values for analysis.

Data Cleaning & Pre-Processing

- Convert Touchscreen to Binary (0/1)

```
df['Touchscreen'] = df['Touchscreen'].apply(lambda x: 1 if x == 'Yes' else 0)
```

Changes 'Yes' to 1 and 'No' to 0.

- Convert IPSpanel to Binary (0/1)

```
df['IPSpanel'] = df['IPSpanel'].apply(lambda x: 1 if x == 'Yes' else 0)
```

Same idea as touchscreen: Encodes whether the screen uses IPS technology.

- Encode RetinaDisplay to Binary (0/1)

```
df['RetinaDisplay'] = df['RetinaDisplay'].apply(lambda x: 1 if x == 'Yes' else 0)
```

Binary format again, indicating presence of Apple's Retina Display.

Data Cleaning & Pre-Processing

- Calculate PPI (Pixels Per Inch)

```
df['PPI'] = ((df['ScreenW']**2 + df['ScreenH']**2)**0.5) / df['Inches']
```

Uses the Pythagorean theorem to calculate screen resolution (diagonal pixels), then divides by screen size in inches.

- Create TotalStorage by Summing Primary and Secondary Storage

```
df['TotalStorage'] = df['PrimaryStorage'] + df['SecondaryStorage']
```

Adds primary and secondary storage values to get total available storage per laptop.

Data Cleaning & Pre-Processing

- Encode Screen Resolution into Numeric Levels

```
df['Screen'] = df['Screen'].apply(lambda x: 1 if x == 'Full HD' else (2 if x == '4K Ultra HD' else 0))
```

Maps screen types to numbers:

0 → other/unknown
1 → Full HD
2 → 4K Ultra HD

- One-Hot Encode Categorical Columns

```
cat_cols = [col for col in ['Company', 'TypeName', 'OS', 'CPU_company', 'PrimaryStorageType', 'SecondaryStorageType', 'GPU_company'] if col in df.columns
df = pd.get_dummies(df, columns=cat_cols, drop_first=True)
```

Converts categorical variables into binary (dummy) variables for machine learning models.

Data Cleaning & Pre-Processing

```
df.dtypes
```

```
Product                    object
Inches                    float64
Ram                         int64
Weight                    float64
Price_euros               float64
Screen                      int64
ScreenW                     int64
ScreenH                     int64
Touchscreen                 int64
IPSpanel                    int64
RetinaDisplay               int64
CPU_freq                  float64
CPU_model                  object
PrimaryStorage              int64
SecondaryStorage            int64
GPU_model                  object
PPI                       float64
TotalStorage                int64
Company_Apple                bool
Company_Asus                 bool
Company_Chuwi                bool
Company_Dell                 bool
Company_Fujitsu              bool
Company_Google               bool
Company_HP                   bool
Company_Huawei               bool
Company_LG                   bool
```

```
Company_Toshiba                  bool
Company_Vero                     bool
Company_Xiaomi                   bool
TypeName_Gaming                  bool
TypeName_Netbook                 bool
TypeName_Notebook                bool
TypeName_Ultrabook               bool
TypeName_Workstation             bool
OS_Chrome OS                     bool
OS_Linux                         bool
OS_Mac OS X                      bool
OS_No OS                         bool
OS_Windows 10                    bool
OS_Windows 10 S                  bool
OS_Windows 7                     bool
OS_macOS                         bool
CPU_company_Intel                bool
CPU_company_Samsung              bool
PrimaryStorageType_HDD           bool
PrimaryStorageType_Hybrid        bool
PrimaryStorageType_SSD           bool
SecondaryStorageType_Hybrid      bool
SecondaryStorageType_No          bool
SecondaryStorageType_SSD         bool
GPU_company_ARM                  bool
GPU_company_Intel                bool
GPU_company_Nvidia               bool
dtype: object
```

Data Cleaning & Pre-Processing

```
df.isnull().sum()
```

```
Company                 0
Product                 0
TypeName                0
Inches                  0
Ram                     0
OS                      0
Weight                  0
Price_euros             0
Screen                  0
ScreenW                 0
ScreenH                 0
Touchscreen             0
IPSpanel                0
RetinaDisplay           0
CPU_company             0
CPU_freq                0
CPU_model               0
PrimaryStorage          0
SecondaryStorage        0
PrimaryStorageType      0
SecondaryStorageType    0
GPU_company             0
GPU_model               0
dtype: int64
```

Here, We checked data is null or not using isnill() function.

Methodology

- Data Collection:

  - We have collected data from google drive.

- Data Pre-processing & Cleaning:

- Converted the Ram column to integer by removing 'GB' from the values.
- Cleaned the Weight column by removing 'kg' and converting it to float.
- Calculated PPI (Pixels Per Inch) based on screen width, height, and inches.
- Combined PrimaryStorage and SecondaryStorage to create TotalStorage.
- Converted categorical columns (Touchscreen, IPSpanel, and RetinaDisplay) into binary format.
- Mapped Screen to numerical values based on screen type (1 for Full HD, 2 for 4K Ultra HD).
- Applied one-hot encoding to categorical columns like Company, TypeName, OS, CPU_company,etc.

Methodology

- Used Algorithm:

- Actual vs Predicted Laptop Prices: Random Forest Regressor algorithm was used. It revealed that laptop prices depend on several factors. This model helps businesses understand which features have the greatest impact on pricing, enabling accurate price prediction.

- Top 10 Important Features: These features were derived from the feature importance scores of the Random Forest Regressor model. Identifying the main factors influencing pricing is crucial for businesses, as it helps them set appropriate product prices and meet customer expectations.

- RAM vs Price: KMeans Clustering algorithm was applied. It segmented laptops into two main categories, revealing a clear relationship between RAM and price. This is useful for businesses to identify which features and price points align with specific customer segments.

Methodology

- Weight vs Price: Isolation Forest algorithm was used for anomaly detection. It identified laptops with unusually high or low prices that deviated from typical ranges. This is essential for businesses to detect and correct pricing errors or incorrect listings in a timely manner.

- Laptop Price Category Classification: Decision Tree Classifier was used to categorize laptops into price ranges (Budget, Midrange, Premium). It showed balanced performance across all categories. This helps businesses understand which features fit into specific price brackets and place products into appropriate categories based on customer needs.

Feature Engineering

Explanation of Feature Selection and Extraction

- We analyzed various features such as Touchscreen, Retina Display, CPU frequency, RAM, Weight, Total Storage, PPI, Screen size, Company, CPU company, GPU company, Type Name, OS, Primary Storage Type, and Secondary Storage Type. By examining these features, we identified the most influential factors affecting laptop pricing. These insights enabled us to build models that understand the relative value of technical specifications in determining price.
  The importance of these features was determined from the model's feature importance scores, enabling businesses to make informed decisions on product pricing and customer satisfaction.

- Using features like RAM, Weight, and Price, we performed segmentation to group laptops into clusters. This revealed customer-preferred configurations and clear patterns in feature-price relationships, enabling more targeted marketing and pricing strategies based on performance tiers.

Feature Engineering

Explanation of Feature Selection and Extraction

- With features like Weight, Screen size, Price, and Total Storage, we detected anomalies in pricing through outlier detection. This helped identify incorrect listings or mispriced products, improving data quality and customer trust.

- By categorizing laptops using key specifications such as RAM, Weight, CPU frequency, Primary Storage, and Secondary Storage, we accurately assigned products into Budget, Midrange, and Premium segments. These classifications supported better assortment planning and feature alignment with customer expectations.

Feature Engineering

Feature Scaling

- We used StandardScaler to scale features like RAM, Weight, and Price because KMeans is a distance-based algorithm. If features have different scales, some features could disproportionately influence the clustering, leading to biased results. Scaling ensures that each feature contributes equally, leading to more balanced and accurate clustering of laptops into distinct categories.

- In Isolation Forest, we used StandardScaler to scale features like Weight,Inches,and TotalStorage. Since the model detects anomalies based on the distance between points, scaling ensures that no feature (like storage or weight) dominates the outlier detection process. This makes the detection of true anomalies more reliable and ensures that the model is equally sensitive to all features.

- Although Decision Trees are not directly affected by feature scaling, we used StandardScaler to scale the features like RAM, Weight, and Storage for consistency. This ensures that all features are on the same scale, which can improve the interpretability of the tree and avoid potential biases when comparing features. Additionally, it helps in faster convergence and more robust model performance.

Feature Engineering

Derived Feature

- PPI (Pixels Per Inch): To calculate screen resolution, which is derived from the screen width and height, and divided by Inches.

  Formula: df['PPI'] = ((df['ScreenW']**2 + df['ScreenH']**2)**0.5) / df['Inches’]

- TotalStorage: The sum of PrimaryStorage and SecondaryStorage, which represents the total storage of the laptop.

  Formula: df['TotalStorage'] = df['PrimaryStorage'] + df['SecondaryStorage’]

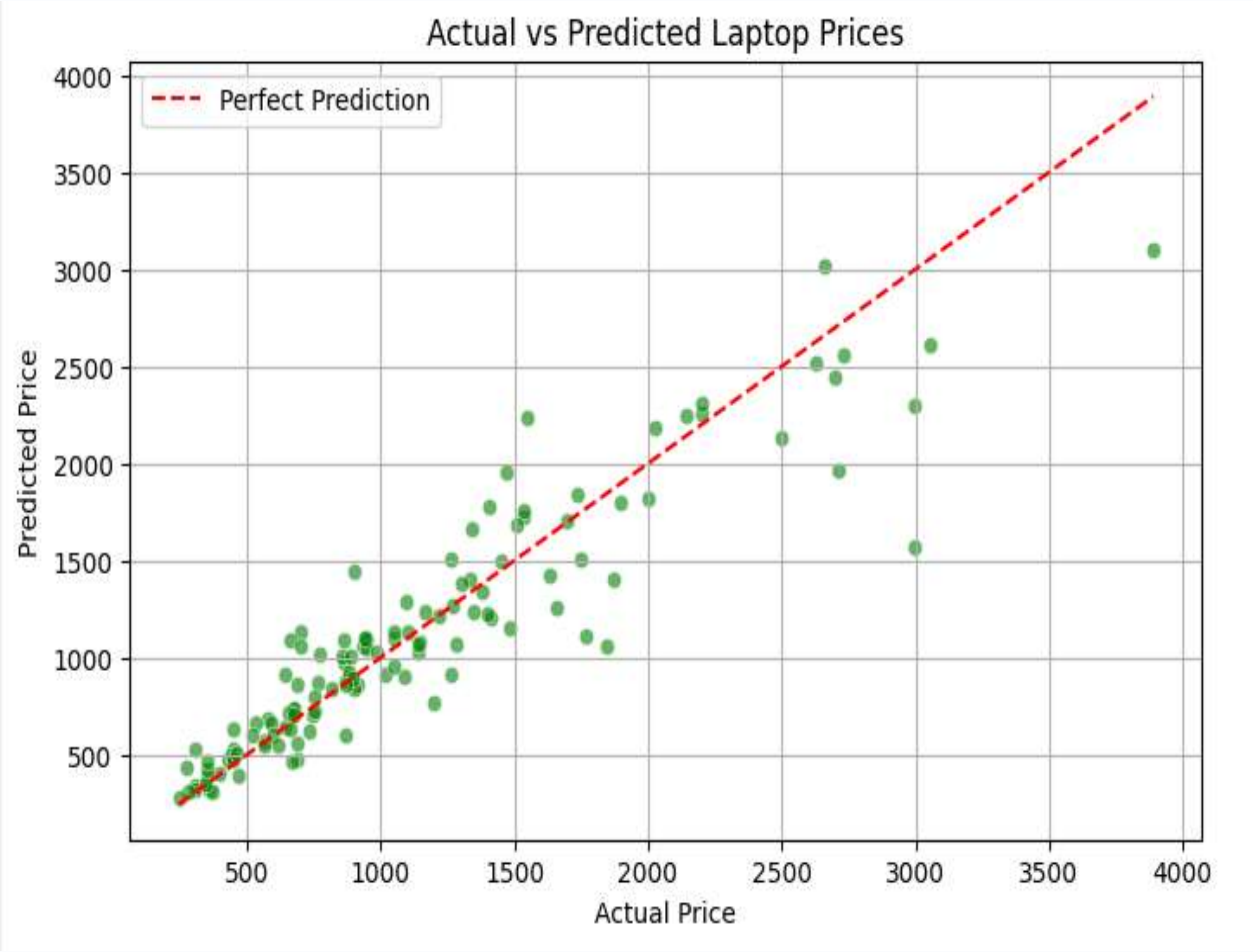
- Anomaly Detection: TotalStorage helps identify outliers with unusually highor low storage, marking them as potential anomalies.

- Price Prediction: PPI improves price prediction by reflecting screen quality. TotalStorage improves accuracy by representing storage capacity, both crucial for pricing.

Model Development

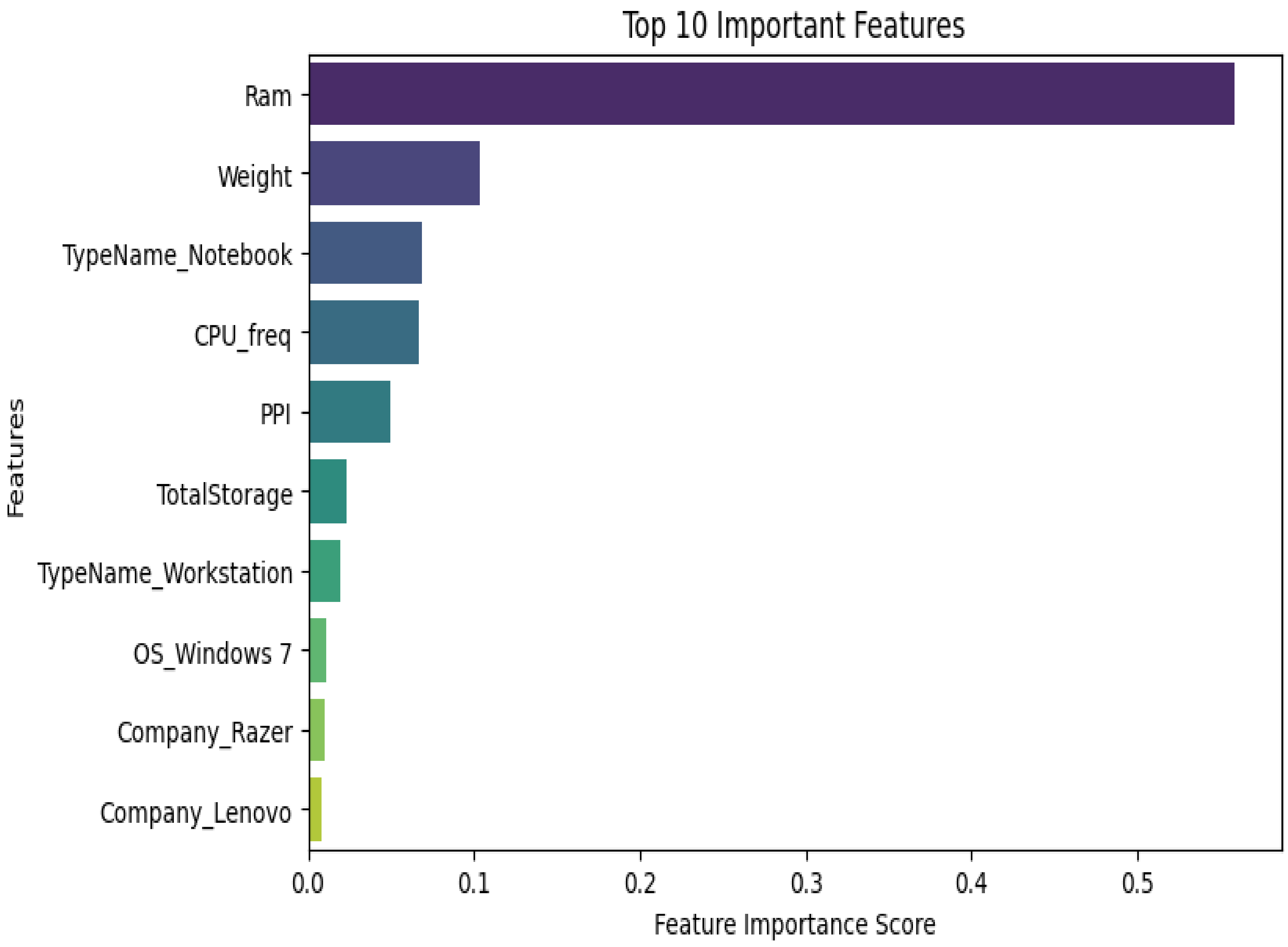
- Through an in-depth analysis of the laptop dataset, we uncovered key pricing drivers, product segmentation insights, and pricing inconsistencies. This exploration revealed how technical features influence pricing, allowing businesses to set competitive and customer-aligned prices. The insights also helped classify laptops into suitable price categories, identify unusual pricing behaviors, and understand customer-focused feature value. By leveraging these data-driven findings, companies can improve pricing strategies, optimize product offerings, and make informed decisions to drive sales and customer satisfaction.

Actual vs Predicted Laptop Prices



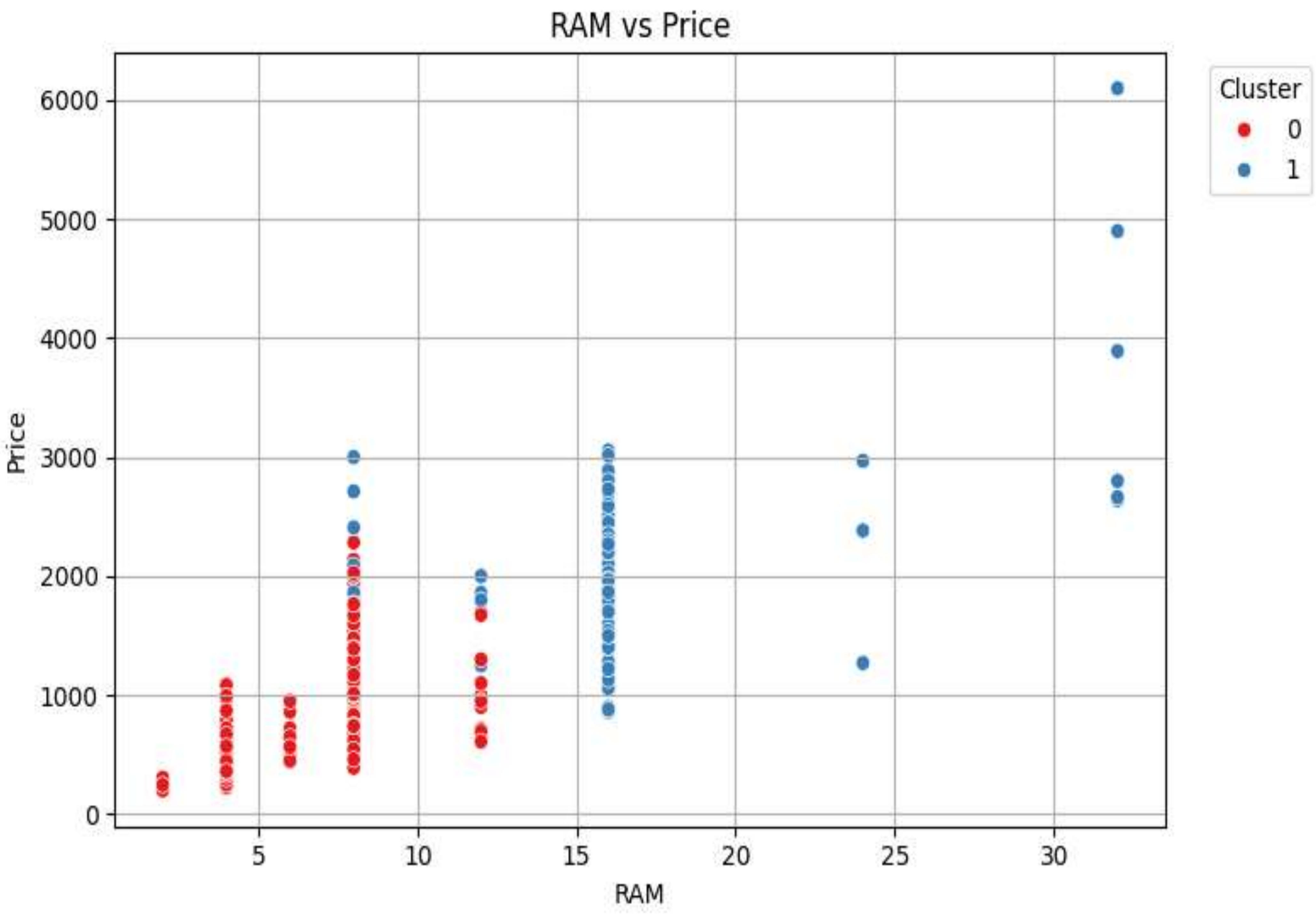
- Actual vs Predicted Laptop Prices: The Random Forest model reveals that laptop prices depend on several factors. This model provides businesses with information on which features are key in influencing laptop prices, enabling accurate pricing.

Top 10 Important Features



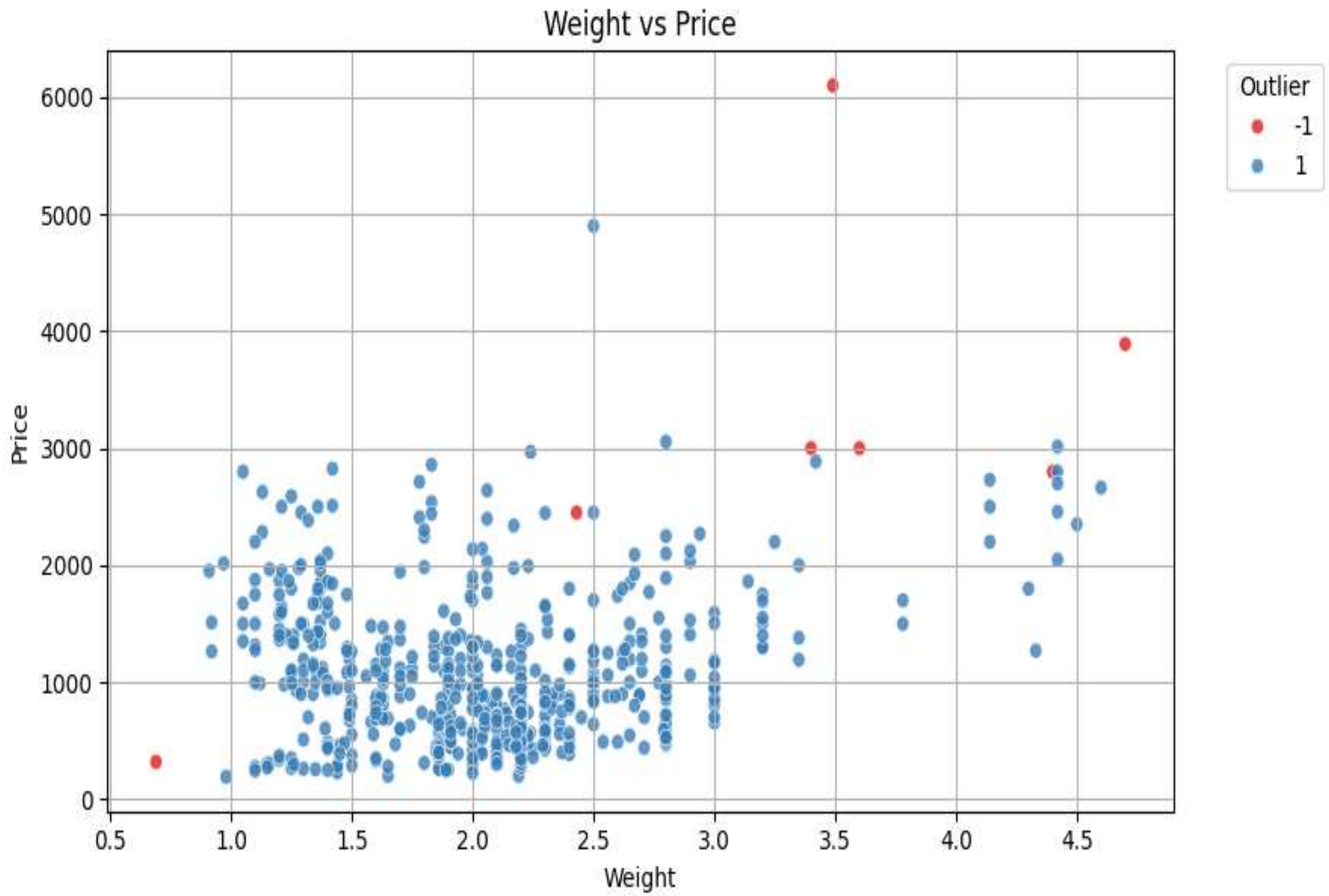
- Top 10 Important Features: Identifying the key factors that influence laptop prices is crucial for businesses, as it helps them determine the prices of their products and meet customer needs.

RAM vs Price



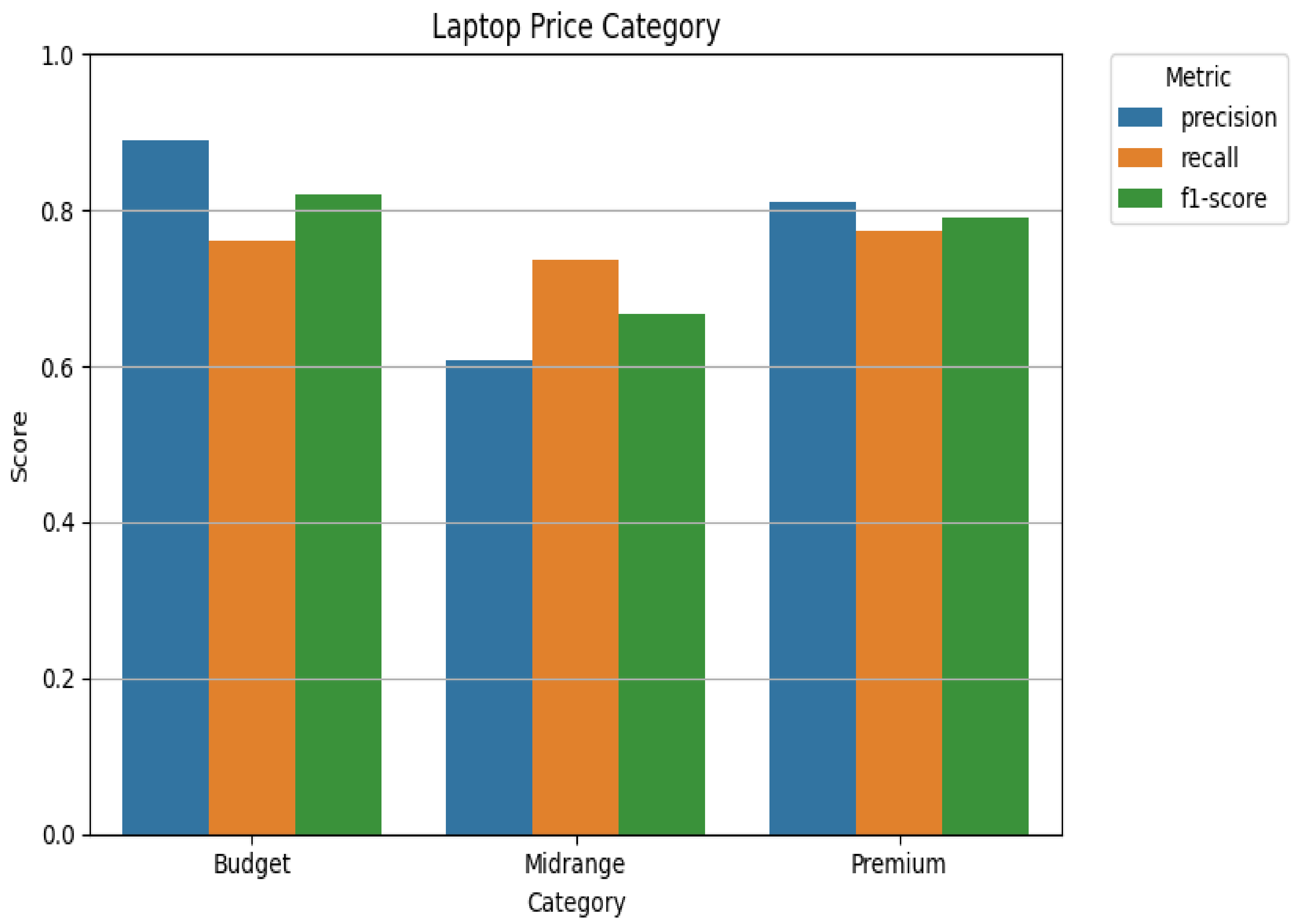
- RAM vs Price: KMeans clustering segmented laptops into two main categories, showing a clear relationship between RAM and price. This is useful for businesses as it helps companies understand which features and price points are most suitable for different customer segments.

Weight vs Price



- Weight vs Price: The Isolation Forest model identified laptops with abnormal prices, significantly higher or lower than normal. This is crucial for businesses as it helps identify products with potential pricing errors or incorrect listings, allowing for timely corrections.

Laptop Price Category



- Laptop Price Category Classification: The Decision Tree model classified laptops into price categories, with balanced performance for each category. This helps businesses understand which features fit specific price categories and enables them to place products in the right category based on customer needs.

Evaluation

- The Random Forest model has an R^2 Score of 0.85, which indicates that the model has accurately captured 85% of the data's variability, meaning the predictions are quite good. The MSE is 75116.81, which shows a moderate error, but overall, the model's performance is quite good.

```
Random Forest:
MSE: 75116.81184432376
R² Score: 0.8485578757096282
```

- The Silhouette Score of 0.5189 indicates moderate clustering quality. It suggests that the clusters have a decent separation, A score closer to 1 would indicate better-defined clusters.

```
Silhouette Score for KMeans with k=2: 0.518916704045863З
```

Evaluation

- The Decision Tree Classifier has an accuracy of 75.81%, which is good for the Budget and Premium classes, but the Midrange class has slightly lower precision (0.61). Overall, the model's performance is good.

```
 Decision Tree Classifier:
Accuracy: 0.7580645161290323

Classification Report:
              precision    recall  f1-score   support

      Budget       0.89      0.76      0.82        42
    Midrange       0.61      0.74      0.67        38
     Premium       0.81      0.77      0.79        44

    accuracy                           0.76       124
   macro avg       0.77      0.76      0.76       124
weighted avg       0.77      0.76      0.76       124
```

Conclusion

Summary of Achievements

- Identification of pricing influencing factors: Internal (such as RAM, processor, etc.) and external (such as market trends) elements were clearly found to affect laptop prices.
- Support in data-driven decisions: Insights obtained from the model contributed to improving strategies related to sales, inventory, and product development.
- Identification of price anomalies: Detection of extremely cheap or expensive products enhanced data accuracy and strengthened customer trust.
- Product segmentation: Clear classification into Budget, Midrange, and Premium categories based on RAM and other features made it easier to target the right customers.
- Effective pricing strategy: Presenting products at the right price not only gave companies a competitive edge in the market but also ensured customer satisfaction.

Thank You